

DKV Unified Engine Dispatch Router
`select_attention_backend(device, batch_size, seq_len)`

Target Backend?

Apple Silicon (MLX)

NVIDIA GPU (CUDA)

CPU / Fallback

MLX Engine (Metal)

mlx.core Metal JIT
Vectorized GEMM

**Fused Low-Rank +
Residual Attention**

In-place Block Store
Unified Memory

CUDA / Triton Engine

Triton Fused Kernel
Autotuned Grid Launch

**Warp-Level Reduction
Tensor Core FP16**

Paged KV Pool &
Exact Residual Cache

PyTorch CPU Fallback

PyTorch Vectorized
Reference SDPA

Full Correctness
Telemetry Verification

Strict Execution
Safety Checks

Unified Output Tensor $O \in \mathbb{R}^{B \times 1 \times H \times d}$ **(Verbatim Recall Preserved)**